WHY WE USE

Statistical Measures

Mean · Median · Mode · Std Dev · Min / Max · Count

BrainerX — Data Science Workshop

Student Performance Dataset

Why Do We Need Statistics?

Imagine you have 1,000 students' scores. You can't read them all.

Statistics give us a way to SUMMARIZE the whole dataset in a few numbers.

Where is the Centre?

Mean, Median, Mode

How Spread Out is it?

Std Dev, Range

What are the Extremes?

Min, Max

How Much Data?

Count

Together, these tell the full story of your data without reading every single row.

01 — The Mean (Average)

What is it?

Add all values together, then divide by how many values there are.

Mean = Sum of all values ÷ Number of values

Example

Scores: 70, 80, 90, 60, 100
→ Sum = 400
→ Mean = 400 ÷ 5 = 80

df.describe() → mean row

Why do we use the Mean?

- It gives ONE number that represents the whole class
- We can compare groups — e.g. did females average higher than males?
- It's the most widely understood measure
- Used in most charts and reports automatically

⚠ **Watch out!**

Outliers (very high/low scores) can pull the mean up or down and give a misleading picture.

02 — The Median (Middle Value)

What is it?

Sort all values from smallest to largest. The median is the one in the middle.

60	70	**80**	90	100

← **Median = 80**

In our dataset, the median is shown as the 50% row in df.describe()

df.describe() → 50% row = Median

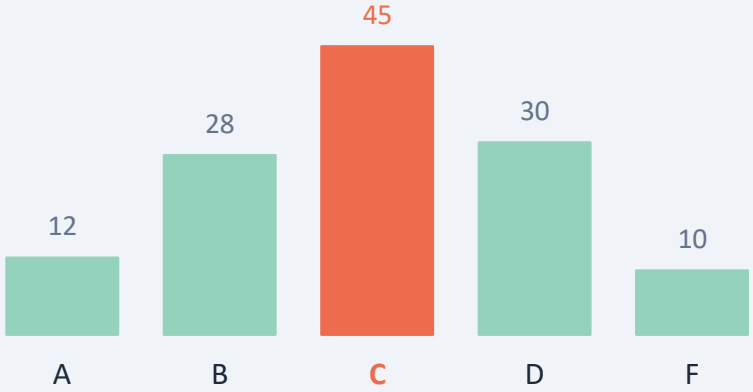
Why do we use the Median?

- It is NOT affected by outliers — more reliable when data is skewed
- e.g. If one student scored 0, the mean drops unfairly — the median stays accurate
- Great for comparing fairly between groups
- When mean ≠ median → the data is skewed — a useful warning sign!

03 — The Mode (Most Frequent)

What is it?

The value that appears most often in the dataset.



← Grade C is the Mode (most students got it)

df['grade'].value_counts() → most frequent = Mode

Why do we use the Mode?

- Shows the most COMMON outcome in your data
- Essential for categorical data (gender, grade, race group)
- Helps identify the 'typical' student — what grade do most students get?
- Not distorted by outliers at all

04 — Standard Deviation (Spread)

What is it?

How far, on average, are students' scores from the mean? A measure of spread.

Class A — Low Std Dev

78 79 80 81 82

Std ≈ 1.5 → everyone near the mean

Class B — High Std Dev

50 65 80 95 100

Std ≈ 20 → very mixed levels

df.describe() → std row

Why do we need Std Dev?

- The mean alone can be misleading — std dev reveals the full picture
- Small std → students are all at a similar level (consistent class)
- Large std → very mixed class — some need extra help, others are advanced
- Helps spot outliers — a score 2+ std devs from the mean is unusual
- Essential for comparing two classes with the same mean but different spreads

05 — Min & Max (The Extremes)

↓ MIN

45

Lowest score in the dataset

The student who struggled the most.

↑ MAX

100

Highest score in the dataset

The top performing student.

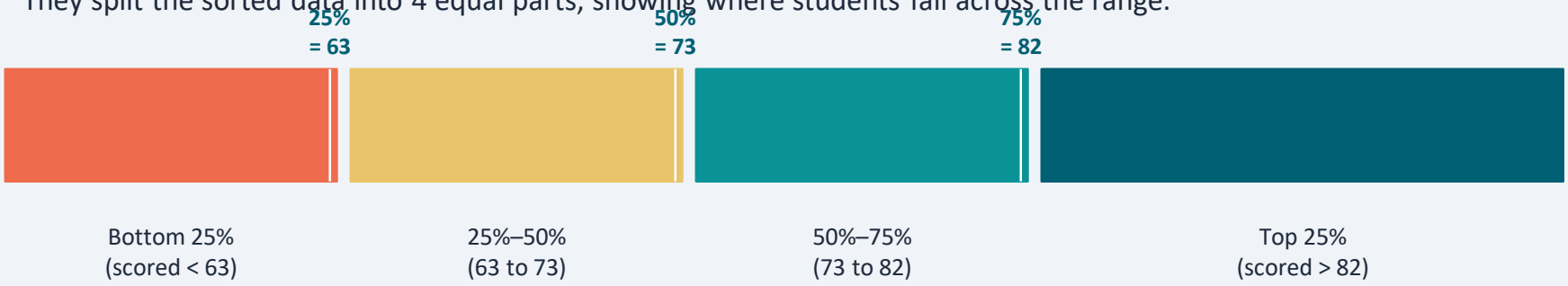
Why? → Min & Max instantly show the RANGE of the data. They reveal if someone scored 0 (data error?) or 100 (top achiever). Combined with the mean, they help you understand if the class is clustered tightly or spread over a wide range.

df.describe() → min and max rows

06 — Percentiles: 25%, 50%, 75%

What are they?

They split the sorted data into 4 equal parts, showing where students fall across the range.



Why do we use Percentiles?

- They tell you how any one student compares to the whole class
- If a student is at the 75th percentile → they scored better than 75% of students
- The gap between 25% and 75% (called the IQR) shows the middle spread of the class
- Used in boxplots — the most popular chart for score comparisons

df.describe() → 25%, 50%, 75% rows

07 — Count (How Many Values?)

1000

students in the dataset

Count = 1000 per column
(if all equal → no missing data)

Why does Count matter?

- It tells you how many non-empty (non-null) values exist in each column
- If count < total rows → you have MISSING DATA — a big red flag!
- Different counts per column means some columns have more gaps than others
- Always the FIRST thing to check before any analysis

df.describe() → count row | df.isnull().sum() for missing values

Putting it All Together — df.describe()

	math_score	reading_score	writing_score
count	1000	1000	1000
mean	66.09	69.17	68.05
std	15.16	14.60	15.20
min	0	17	10
25%	57.00	59.00	57.75
50%	66.00	70.00	69.00
75%	77.00	79.00	79.00
max	100	100	100

count
How many non-empty values

mean
Average score

std
How spread out the scores are

min / max
Lowest and highest score

25% / 50% / 75%
Quartiles — class distribution

Run df.describe() after loading any dataset — it gives you all 7 measures instantly!

When to Use Which Measure?

MEAN

Use when:

Normal data, no extreme outliers

Example:

Average score of the class

Avoid when:

When there are extreme outliers

MEDIAN

Use when:

Skewed data or outliers present

Example:

When one student scored 0 unfairly

Avoid when:

When you need a precise average

MODE

Use when:

Categorical or repeated values

Example:

Most common grade in the class

Avoid when:

For precise numerical analysis

When in doubt → report all three. The differences between them tell the most interesting story.

QUICK RECAP

Every Measure Tells a Different Story

Count → Are there missing values?

Mean → What is the typical score?

Median → What is the middle score (no outlier bias)?

Mode → What is the most common grade?

Std Dev → How mixed are the students' levels?

Min / Max → What is the full range of scores?

25% / 75% → Where do the lower and upper halves start?

Run df.describe() → you get ALL of these in one single command